WORKED SOLUTION

STEP I 2007, Question 11

Mechanics • Projectile motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

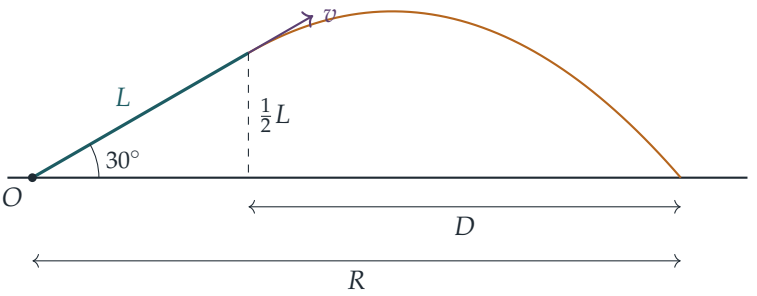
step.maths.org/questions/07-s1-q11

The key idea

Split the motion into two stages. Inside the smooth tube the only force along the tube is the component of gravity, so energy (or constant-acceleration formulae) gives the exit speed. After that it is an ordinary projectile, launched at 30° from a point a height $\frac{1}{2}L$ above the finish level. Eliminating the flight time gives the quadratic for D; differentiating it *implicitly* gives $\mathrm{d}D/\mathrm{d}L$ without ever solving for D.

Solution

We model the particle as a point mass, the tube as smooth (so the normal reaction does no work and has no component along the tube), and ignore air resistance.



Inside the tube

Resolving along the tube, the only force component is $mg\sin 30^\circ = \frac{1}{2}mg$ down the slope, so the deceleration is $\frac{1}{2}g$. Over distance L,

$$v^2 = u^2 - 2\cdot\tfrac{1}{2}g\cdot L = u^2 - gL,$$

where v is the exit speed. (We need $u^2 > gL$ for the particle to leave the tube at all.) The top of the tube is a height $L\sin 30^\circ = \frac{1}{2}L$ above the lower end and a horizontal distance

$L\cos 30^\circ = \frac{\sqrt{3}}{2}L$ from it.

The projectile stage

The particle leaves along the tube, so its velocity has horizontal component $\frac{\sqrt{3}}{2}v$ and vertical component $\frac{1}{2}v$ upwards. Measuring from the exit point, after time t

$$x = \tfrac{\sqrt{3}}{2}vt, \qquad y = \tfrac{1}{2}vt - \tfrac{1}{2}gt^2.$$

It returns to the level of the lower end when $y = -\frac{1}{2}L$, at which moment $x = D$. From the first equation $t = \dfrac{2D}{\sqrt{3}\,v}$, and substituting into the second,

$$\frac{D}{\sqrt{3}} - \frac{2gD^2}{3v^2} = -\frac{L}{2}.$$

Multiplying through by $-6v^2$,

$$4gD^2 - 2\sqrt{3}\,v^2 D - 3Lv^2 = 0,$$

and putting $v^2 = u^2 - gL$,

$$4gD^2 - 2\sqrt{3}\,(u^2 - gL)D - 3L(u^2 - gL) = 0,$$

which is the required equation. (Of its two roots, D is the positive one: the product of the roots is $-3Lv^2/(4g) < 0$.)

The derivative

For fixed u, D depends on L, so we differentiate the equation with respect to L, using the product rule on each term containing both:

$$8gD\frac{\mathrm{d}D}{\mathrm{d}L} - 2\sqrt{3}(u^2 - gL)\frac{\mathrm{d}D}{\mathrm{d}L} + 2\sqrt{3}\,gD - 3(u^2 - gL) + 3gL = 0.$$

The last two terms combine to $-3(u^2 - 2gL)$. Collecting the $\mathrm{d}D/\mathrm{d}L$ terms,

$$\left(8gD - 2\sqrt{3}(u^2 - gL)\right)\frac{\mathrm{d}D}{\mathrm{d}L} = -\left(2\sqrt{3}\,gD - 3(u^2 - 2gL)\right),$$

so

$$\frac{\mathrm{d}D}{\mathrm{d}L} = -\frac{2\sqrt{3}\,gD - 3(u^2 - 2gL)}{8gD - 2\sqrt{3}(u^2 - gL)},$$

as required.

Ask yourself Why does the $-3L(u^2 - gL)$ term produce two pieces?

Because L appears twice in it: differentiating $-3Lu^2 + 3gL^2$ gives $-3u^2 + 6gL$, which is $-3(u^2 - gL) + 3gL$. Forgetting the L inside the bracket is the easiest way to miss the $2gL$ in the answer.

Stationary R

The final horizontal displacement from the lower end is the horizontal run of the tube plus D:

$$R = \tfrac{\sqrt{3}}{2}L + D, \qquad \text{so} \qquad \frac{\mathrm{d}R}{\mathrm{d}L} = \frac{\sqrt{3}}{2} + \frac{\mathrm{d}D}{\mathrm{d}L}.$$

Hence $\mathrm{d}R/\mathrm{d}L = 0$ exactly when $\mathrm{d}D/\mathrm{d}L = -\frac{\sqrt{3}}{2}$, that is (multiplying up by the denominator, which we check below is non-zero)

$$2\sqrt{3}\,gD - 3(u^2 - 2gL) = \tfrac{\sqrt{3}}{2}\big(8gD - 2\sqrt{3}(u^2 - gL)\big) = 4\sqrt{3}\,gD - 3(u^2 - gL).$$

The $3u^2$ terms cancel, leaving $6gL - 2\sqrt{3}\,gD = 3gL$, so $2\sqrt{3}\,gD = 3gL$, i.e.

$$2D = L\sqrt{3}.$$

Every step is reversible, so $\mathrm{d}R/\mathrm{d}L = 0$ when $2D = L\sqrt{3}$, as required.

To find R, substitute $D = \frac{\sqrt{3}}{2}L$ into the quadratic:

$$4g \cdot \tfrac{3}{4}L^2 - 2\sqrt{3}(u^2 - gL) \cdot \tfrac{\sqrt{3}}{2}L - 3L(u^2 - gL) = 3gL^2 - 6L(u^2 - gL) = 0.$$

Since $L > 0$, this gives $gL = 2(u^2 - gL)$, so $L = \dfrac{2u^2}{3g}$. (Then $v^2 = u^2 - gL = \frac{1}{3}u^2 > 0$, so the particle does leave the tube, and the denominator $8gD - 2\sqrt{3}v^2 = \frac{8\sqrt{3}}{3}u^2 - \frac{2\sqrt{3}}{3}u^2 = 2\sqrt{3}\,u^2$ is non-zero.) Finally

$$R = \tfrac{\sqrt{3}}{2}L + D = \sqrt{3}\,L = \frac{2\sqrt{3}\,u^2}{3g} = \frac{2u^2}{\sqrt{3}\,g}.$$

Checks

- With $L = 0$ the quadratic gives $D = \sqrt{3}u^2/(2g)$, the familiar range $u^2 \sin 60^\circ / g$ for a launch at 30°.
- With $L = u^2/g$ the particle just reaches the top with $v = 0$, so $D = 0$ and $R = \frac{\sqrt{3}}{2}u^2/g$ again. The stationary value $2u^2/(\sqrt{3}g) \approx 1.155\,u^2/g$ lies between these and above both, so it is a maximum.
- Numerically, with $u = g = 1$, scanning L over $[0, 1]$ gives a largest R of 1.1547 at $L \approx 0.6667$, matching $2/\sqrt{3}$ and $L = \frac{2}{3}$; stepping the motion directly also reproduces the quadratic for D.

Common slips

- Taking the deceleration in the tube to be g rather than $g \sin 30^\circ = \frac{1}{2}g$.
- Setting $y = 0$ instead of $y = -\frac{1}{2}L$: the particle lands below its launch point.
- Writing $R = D$, forgetting the horizontal length $\frac{\sqrt{3}}{2}L$ of the tube itself.

Beyond STEP

Differentiating an equation $F(D, L) = 0$ without solving it is implicit differentiation, and in first-year analysis it becomes the implicit function theorem: $\mathrm{d}D/\mathrm{d}L = -F_L/F_D$, valid wherever $F_D \neq 0$, which is exactly the denominator we checked. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for the mechanics.

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